Appendix (Multi-Agent Dual-Path Verification Framework for Scalable Graph Reasoning)

A Prompt Templates

A.1 Router Agent

Prompt Template

```
system: "You are a query router for a hybrid neural-
    symbolic graph reasoning system."
user_template: |
  Query: {query}
  TaskType: {task_type}
  GraphStats: {graph_stats}

  Decide route:
  - symbolic: pick this when the query asks for a
    computation that has a precise procedural answer
     (path, flow, matching, substructure, traversal,
     counting). The system will generate and
    execute code.
  - neural: pick this when the query asks for an
    interpretation, preference, or attribute-based
    judgment that does not reduce to a single
    procedure. The system will reason directly.

  Output JSON:
  {{
    "route": "symbolic" or "neural",
    "why": "Brief reasoning",
    "need_decomposition": true/false,
    "decomposition": {{
      "seed_nodes": [],
      "hop": {default_hop},
      "node_budget": {node_budget},
      "edge_budget": {edge_budget}
    }}
  }}
```

A.2 Planner Agent

Prompt Template

```
system: "You are a graph task planner. Analyze the
    problem and outline logical steps for an
    implementation."
user_template: |
  Task: {query}
  TaskType: {task_type}
  GraphStats: {graph_stats}
  GraphDataHint: {graph_hint}

  Provide a high-level execution plan:
  - algorithm_family: the category of algorithm (e.g
    ., BFS, flow, backtracking, isomorphism).
  - key_steps: logical steps to solve the problem
    programmatically.
  - defensive_checks: edge cases to handle (
    disconnected graph, missing nodes, type
    mismatch).

  Output JSON:
  {{
    "algorithm_family": "...",
    "key_steps": ["..."],
    "defensive_checks": ["..."]
  }}
```

A.3 Coder Agent

Prompt Template

```
system: "You are an expert Python programmer
    specializing in NetworkX. Write robust, fault-
    tolerant code."
user_template: |
  Task: {query}
  TaskType: {task_type}
  Plan: {plan_json}

  Graph Input Data:
  {graph_payload_hint}

  Constraints:
  1. Use only standard Python libraries and networkx.

  2. Do not install new packages.
  3. Print only the final answer. Do not print debug
     info.
  4. Wrap dangerous code in try/except blocks and
    print a safe fallback if it fails.

  If this is a retry, fix the error:
  {error_feedback}

  Write the complete Python code in a single ‘‘‘
    python‘‘‘ block.
```

A.4 Reasoner Agent

Prompt Template

```
system: "You are a logical graph reasoner. Solve the
     problem step-by-step using semantic reasoning
    ."
user_template: |
  Task: {query}
  TaskType: {task_type}
  GraphSummary: {graph_hint}

  Perform chain-of-thought reasoning and provide the
     final answer.
```

```
  Answer:
```

A.5 Critic Agent

Prompt Template

```
system: "You are a strict execution verifier.
     Compare two outputs for logical equivalence."
consistency_template: |
  Query: {query}
  OutputA: {out_a}
  OutputB: {out_b}

  Compare the two outputs.
  1. If both are effectively the same (e.g., "Yes"
     vs "True", "0-1" vs "1-0"), output consistent=
     true.
  2. If one output implies failure (e.g., "No", "
     Error", "No path") and the other provides a
     specific concrete solution, prefer the concrete
      solution unless it is obviously hallucinated.
  3. If they are logically contradictory (e.g., "
     Node 1" vs "Node 2"), explain the conflict.

  Output JSON:
  {{
    "consistent": true/false,
    "why": "Brief explanation",
    "resolution": "pick_a" or "pick_b" or "
     need_tiebreaker",
    "tiebreaker_hint": "Explain what specific check
     is needed to verify the truth."
  }}
```

B Per-Benchmark Detailed Results

B.1 GraphWiz

Table 1: Performance comparison on GraphWiz benchmark in terms of accuracy (%).

Task	GPT	GraphTeam	DuVerG
Cycle	82.0	**100.0**	99.2
Connect	64.8	86.5	**100.0**
Bipartite	67.3	96.3	**100.0**
Topology	**100.0**	**100.0**	95.0
Shortest	8.5	95.0	**100.0**
Triangle	10.3	**94.0**	89.2
Flow	19.3	94.0	**97.5**
Hamilton	36.3	32.5	**99.2**
Subgraph	40.3	99.3	**100.0**
Avg	47.6	88.6	**97.8**

B.2 NLGraph

Table 2: Performance comparison on NLGraph benchmark in terms of accuracy (%).

Task	GPT	GraphTeam	DuVerG
Connect	72.0	97.0	**100.0**
Cycle	52.9	**100.0**	**100.0**
Topology	20.0	94.8	**97.8**
SP	35.9	98.4	**100.0**
MaxFlow	98.3	**100.0**	98.3
Bipartite	29.8	**100.0**	98.8
Hamilton	22.4	**100.0**	**100.0**
GNN	0	97.4	**100.0**
Avg	51.4	97.8	**99.5**

B.3 Talk Like a Graph

Table 3: Performance comparison on Talk Like a Graph benchmark in terms of accuracy (%).

Task	GPT	GraphTeam	DuVerG
NCount	**100.0**	**100.0**	99.3
ECount	45.6	99.2	**100.0**
EExist	87.0	61.0	**100.0**
NDeg	54.6	98.6	**99.3**
CNodes	50.2	99.2	**100.0**
Cycle	92.6	99.4	**100.0**
SP	45.6	99.1	**100.0**
TriCount	14.4	99.2	**99.3**
DiscNodes	41.3	54.0	**84.7**
MaxFlow	36.7	77.3	**98.0**
NClass	51.3	74.0	**100.0**
Reach	97.3	98.0	**100.0**
Avg	59.7	88.3	**98.4**

B.4 GraphInstruct

Table 4: Performance comparison on GraphInstruct benchmark in terms of accuracy (%).

Task	GPT	GraphTeam	DuVerG
Neighbor	98.0	**100.0**	99.0
Bipartite	89.7	99.0	**100.0**
Edge	90.7	**99.3**	94.7
Connectivity	92.2	**100.0**	99.7
Degree	98.0	**100.0**	99.7
DFS	89.3	95.7	**97.0**
Predecessor	47.4	98.0	**99.7**
TopoSort	59.0	98.3	**99.7**
ConnComp	39.3	94.7	**99.7**
CommonNbr	49.0	**100.0**	99.0
Hamilton	33.7	97.7	**100.0**
Jaccard	17.0	**100.0**	99.0
SP	57.3	**100.0**	**100.0**
Diameter	23.0	**100.0**	99.7
PageRank	15.3	16.3	**91.7**
MaxFlow	15.0	99.3	**100.0**
MST	1.0	0	**100.0**
Avg	53.7	88.1	**98.7**

B.5 GraphArena

Table 5: Performance comparison on GraphArena benchmark in terms of accuracy (%).

Task	GPT	GraphTeam	DuVerG
Connect	56.1	**100.0**	95.1
Diameter	32.3	0.0	**95.0**
Distance	61.8	92.4	**99.3**
GED	20.0	3.1	**38.0**
MCP	29.6	98.0	**99.7**
MCS	50.7	12.7	**86.0**
MIS	26.9	68.3	**90.6**
MVC	14.7	9.7	**49.3**
Neighbor	82.0	98.8	**99.8**
TSP	14.3	23.0	**63.7**
Avg	39.2	50.6	**83.8**

B.6 OGB Benchmarks

Table 6: Performance comparison on OGB benchmarks in terms of accuracy (%).

Dataset / Task	GPT	GraphTeam	DuVerG
ogbn-arxiv			
TriCount	13.5	99.2	**100.0**
ClustCoef	14.6	83.5	**100.0**
3-Reach	44.2	**99.3**	98.1
PageRank	15.5	30.3	**76.2**
Avg	22.4	79.0	**93.5**
ogbl-ppa			
TriCount	3.2	**100.0**	**100.0**
ClustCoef	3.8	88.0	**100.0**
3-Reach	30.9	**85.4**	83.8
PageRank	3.2	21.1	**47.8**
Avg	10.5	74.6	**82.6**